

Final Project Report Template

1. Introduction

1.1. Project overviews

This project, "**Analysing Mental Health in Student Ecosystem**", is the Skillwallet "Data Analytics with Tableau" capstone, completed individually by Atharv Bhosale.

It analyses the Personalized Psychological Intervention dataset (1,000 student records, 16 fields) in Tableau to see how sleep, study hours, attendance, stress, depression and heart rate variability relate to students' mental health scores and risk levels. The output is a set of Tableau worksheets, an interactive dashboard and a Tableau Story that counsellors and faculty mentors can use to understand student well-being.

1.2. Objectives

- Understand the structure, quality and distribution of the student mental health dataset.
- Frame clear business questions that a counsellor or mentor would want answered.
- Build Tableau visualizations that answer each question.
- Combine them into a dashboard with key KPIs: average mental health score, sleep duration, depression score and attendance rate.
- Present the findings as a Tableau Story, with a clear conclusion on whether any single factor predicts risk.

2. Project Initialization and Planning Phase

2.1. Define Problem Statement

Two problem statements were defined, each in the format "I am / trying to / but / because / which makes me feel":

- **PS-1:** a college counsellor who wants to identify at-risk students early.
- **PS-2:** a faculty mentor who wants to understand which student factors need attention.

2.2. Project Proposal (Proposed Solution)

The proposed solution is a set of Tableau dashboards and a Story built on the 1,000-record dataset, so that counsellors and mentors can see patterns in stress, depression, sleep, attendance and risk level across gender, grades and study habits without reading raw data.

Tools: Tableau Public, Excel/CSV data source, VS Code, Git, and a Flask website (ePortfolio) to host the dashboards.

2.3. Initial Project Planning

The work was planned as 16 user stories (USN-1 to USN-16) across 7 epics, each estimated at 2 to 5 story points, delivered over three sprints:

Sprint	Dates	Focus
Sprint-1	14 Sep to 20 Sep 2026	Connected the dataset (1,000 rows, 16 fields), validated field types and explored the risk distribution.
Sprint-2	21 Sep to 27 Sep 2026	Built the analysis sheets (risk split, sleep vs stress, metric averages, age and journal checks) and set the colour palette.
Sprint-3	28 Sep to 4 Oct 2026	Built the KPI summary, Gender and Academic analysis sheets, dashboards with filters and the Metric parameter, then the Story, web integration, performance testing and demo.

3. Data Collection and Preprocessing Phase

3.1. Data Collection Plan and Raw Data Sources Identified

Item	Detail
Source	Personalized_Psychological_Intervention_Dataset.xlsx (CSV export has the same data)
Size	1,000 rows, 16 fields, about 129 KB
Location / access	Skillwallet project resources, which are open to only course participants .

3.2. Data Quality Report

The dataset was profiled before analysis:

- No missing values and no duplicate rows; Student_ID is unique and there is no stray whitespace.
- 14 outliers by the 1.5 x IQR rule: 12 in Sleep_Duration_Hours (2.84 to 10.37 hours) and 2 in Heart_Rate_Variability (36.6 and 99.21). They were kept because the values are plausible.
- Emotional_Journal has only 10 distinct sentences.
- The intervention strategy field has 8 values and does not follow the risk level.
- The column Attendance_Rate (%) has special characters in its name.

3.3. Data Exploration and Preprocessing

- Gender: Male 341, Female 331, Other 328. Age range 18 to 25.
- Risk level: Low 414, Medium 347, High 239 (about 24% High).
- No rows needed to be removed or imputed, and outliers were retained.

Very little preprocessing was needed. The data went into Tableau as it was, with a few calculated fields added (see section 7.2).

4. Data Visualization

4.1. Framing Business Questions

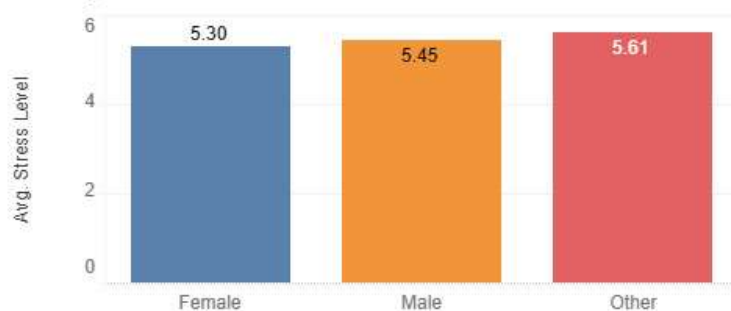
Eight business questions guided the visualizations:

#	Business question
1	How does average stress differ by gender?
2	How does average depression score differ by gender?
3	How does heart rate variability differ by gender?
4	What share of each gender is at High risk?
5	How are academic grades distributed?
6	Does sleep duration relate to stress?
7	Do study hours relate to academic score?
8	How does academic score vary across study-hour buckets?

4.2. Developing Visualizations

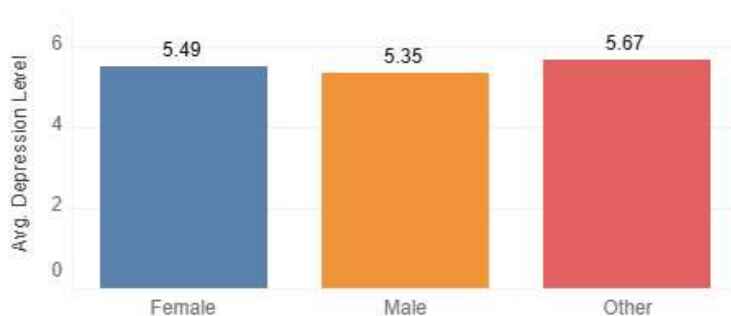
- **Stress by gender:** average stress is 5.30 (Female), 5.45 (Male) and 5.61 (Other).

Average stress is similar across genders (5.3 to 5.6)



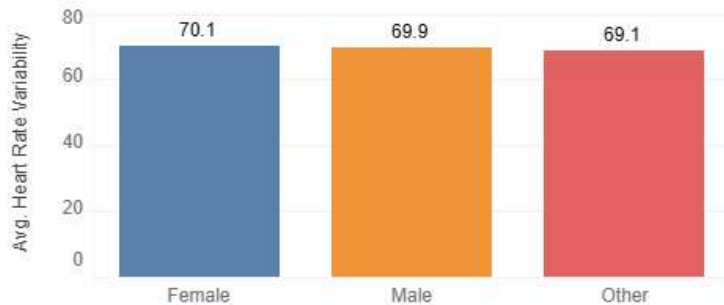
- **Depression by gender:** 5.49 (Female), 5.35 (Male) and 5.67 (Other).

Average depression is similar across genders (5.3 to 5.7)



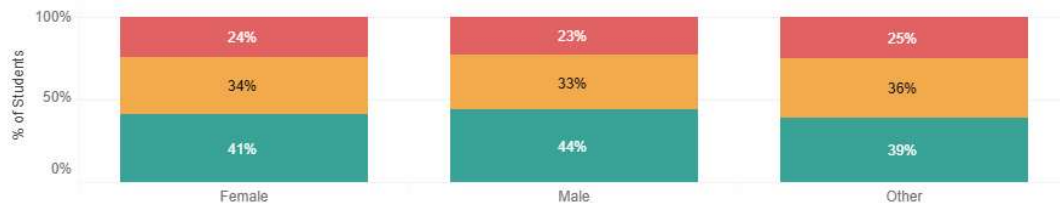
- **Heart rate variability:** 70.1 (Female), 69.9 (Male) and 69.1 (Other).

Heart rate variability is nearly identical across genders (69 to 70)



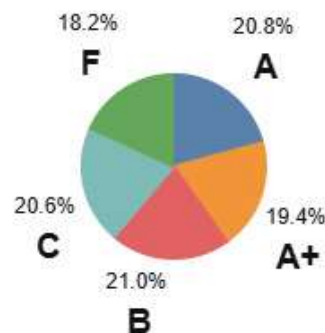
- **Risk by gender:** High risk is 23% to 25% in every gender.

Risk mix is nearly the same across genders (High risk: 23% to 25%)



- **Grades:** each grade band holds 18.2% to 21.0% of students.

Grades are spread almost evenly: no grade dominates (18% to 21%)

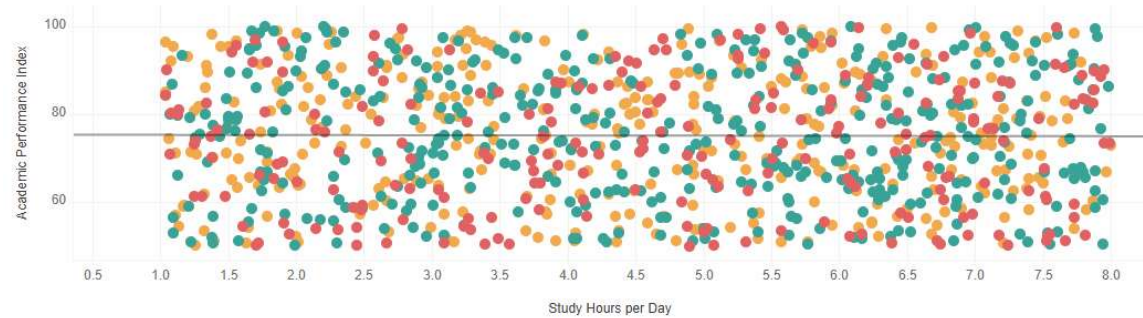


- **Sleep vs stress:** R-squared is 0.0006, so there is effectively no relationship.



- **Study hours vs academic score:** R-squared is about 0 ($p = 0.84$).

Study hours don't predict academic performance ($R^2 \approx 0$, $p = 0.84$)



- **Academic score by study-hour bucket:** average academic score is 73 to 76 in every bucket.

5. Dashboard

5.1. Dashboard Design File

Dashboard 1 brings the KPIs and worksheets onto one page. KPI values: average mental health score 5.45, average sleep 6.49 hours, average depression 5.50 and average attendance 79.96%.

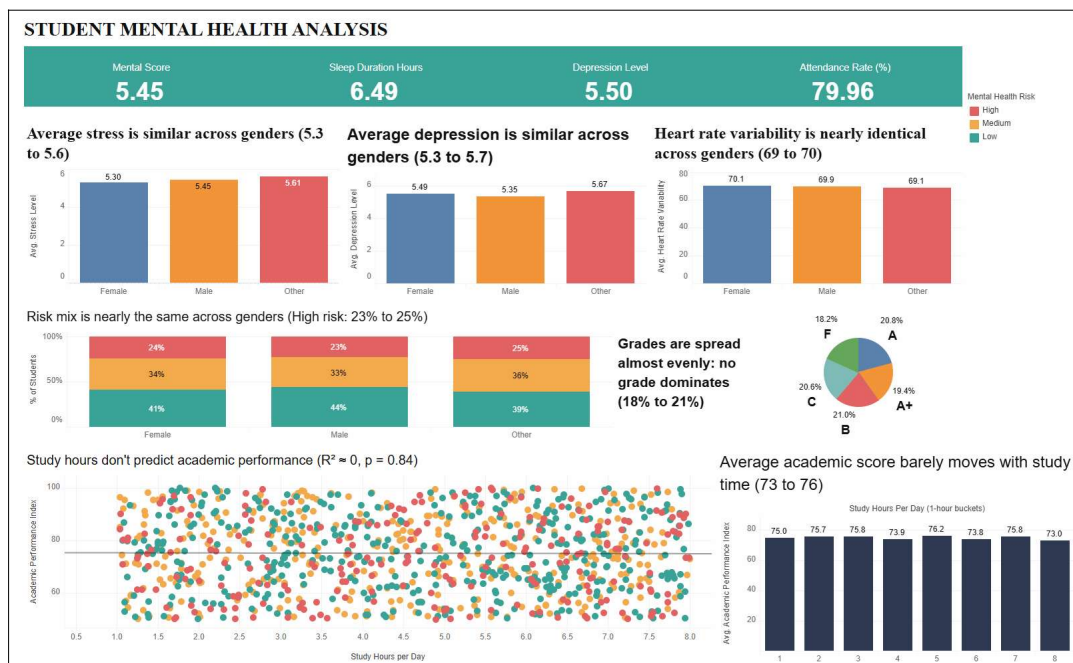
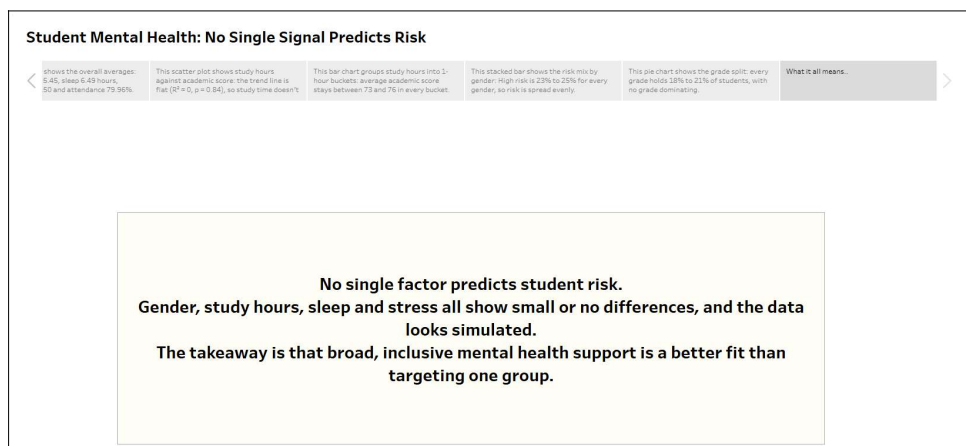


Tableau Public link: [Click here to Dashboard](#)

6. Report

6.1. Story Design File

The Tableau Story, "Student Mental Health: No Single Signal Predicts Risk", walks through the dashboard findings point by point and ends with the conclusion that no single factor predicts risk.



Story link: [Click here](#)

7. Performance Testing

7.1 Utilization of Data filters

Number of data filters used: **2** (Gender and Age),

applied to all sheets that use the data source. Age is a slider from 18 to 25.

Parameters: **1**. The Metric parameter drives the dropdown on the box plot.

7.2 No of Calculation Field

Number of calculated fields: **5**

Calculated field	Purpose
Selected Metric	CASE statement driven by the Metric parameter
Average Mental Score	KPI field: average of the Stress, Anxiety and Depression scores (about 5.45)
High Risk %	Share of High-risk students
Total Students	COUNTD of Student ID
Grade	We used 10-point bands: A+ is 90 and above, A 80 to 89, B 70 to 79, C 60 to 69, and F below 60

7.3 No of Visualization

Number of visualizations (sheets): **21**

The dashboard alone shows 7 charts plus the KPI tiles.

Dashboards: **1** Stories: **1**

8. Conclusion/Observation

- No single factor in this dataset predicts risk: stress, depression, HRV and high-risk share are all close across genders, and study hours and sleep show almost no relationship with academic score or stress.
- The data looks synthetic: grade bands are almost equal, Emotional_Journal has only 10 distinct sentences, and intervention strategy does not follow risk level.
- Because risk is spread evenly, broad inclusive support is a better fit than targeting one group.

9. Future Scope

- Host the dashboards on a Flask website with the Tableau Public embeds.
- Repeat the analysis on real student data, with a time dimension to track change over a semester.
- Add predictive modelling for risk level once real data is available.
- Add alerts or filters so counsellors can drill down to individual cohorts.

10. Appendix

10.1. Source Code(if any)

The ePortfolio website is a small Flask app. The file **app.py** defines the four page routes:

```
from flask import Flask, render_template

app = Flask(__name__)

@app.route('/')
def home():
    return render_template('index.html', page='home')

@app.route('/dashboard')
def dashboard():
    return render_template('dashboard.html', page='dashboard')

@app.route('/story')
def story():
    return render_template('story.html', page='story')

@app.route('/portfolio')
def portfolio():
    return render_template('portfolio.html', page='portfolio')

if __name__ == '__main__':
    app.run(debug=True)
```

The pages are in the **templates** folder (base.html, index.html, dashboard.html, story.html, portfolio.html) and the styling is in **static/style.css**. The dashboard and story pages hold the Tableau Public embed codes.

10.2. GitHub & Project Demo Link

GitHub: [**GitHub repository link**](#)

Project demo video: [**Demo Video**](#)

Tableau Public dashboard: [**Dashboard**](#)

Tableau Public story: [**Story Link**](#)